Introduction to Limits

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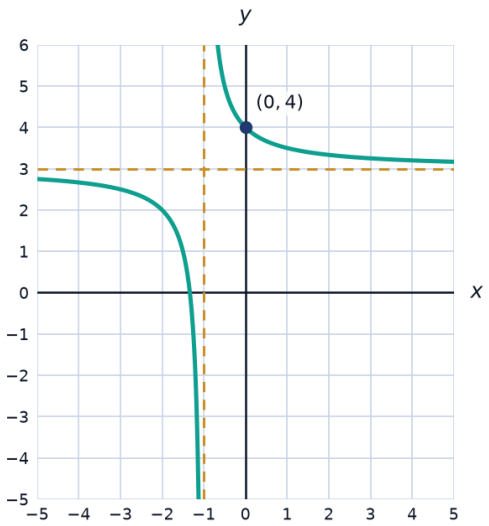
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits**

Evaluate $\lim_{x \to 2} \frac{x^2 - 4}{x - 2}$ by first simplifying the expression.

2 **[Foundational]** • **Target: Evaluating limits**

Evaluate by direct substitution where possible:

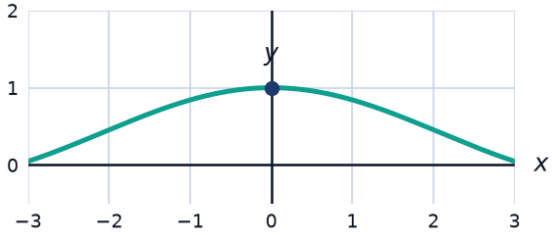


a $\lim_{x \to 3} (x^2 + 2x - 1)$

b $\lim_{x \to 1} \frac{x + 5}{x + 1}$

3 **[Foundational]** • **Target: Evaluating limits**

Using a table of values near $x = 0$ (in radians), estimate $\lim_{x \to 0} \frac{\sin x}{x}$.



4 **[Foundational]** • **Target: One-sided limits and limits at infinity**

Evaluate each limit at infinity:

a $\lim_{x \to \infty} \frac{3x^2 + 5}{x^2 - 2}$

b $\lim_{x \to \infty} \frac{2x + 1}{x^2 + 4}$

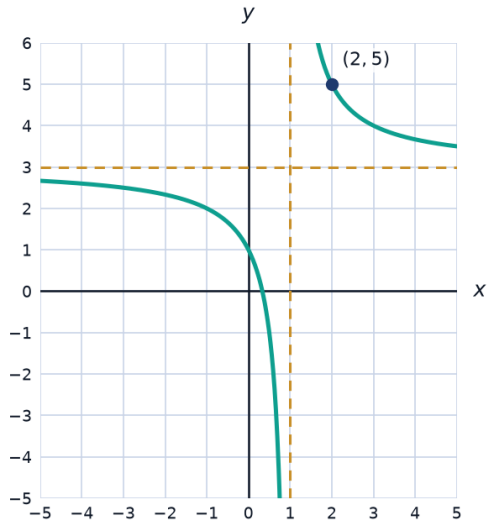
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5 [Foundational] • Target: One-sided limits and limits at infinity

The graph of $f(x) = \frac{1}{x-2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.

6 [Foundational] • Target: More limit computations

Evaluate $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$ by rationalizing the numerator.



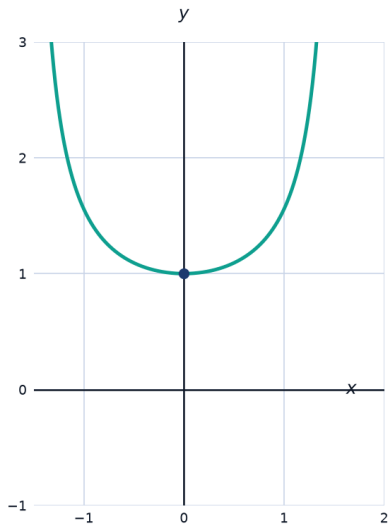
7 [Foundational] • Target: More limit computations

Evaluate $\lim_{x \to -1} \frac{x^3 + 1}{x + 1}$ by factoring (using the sum-of-cubes formula).

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More limit computations

Evaluate $\lim_{x \to 0} \frac{\tan x}{x}$, using $\lim_{x \to 0} \frac{\sin x}{x} = 1$ and $\tan x = \frac{\sin x}{\cos x}$.



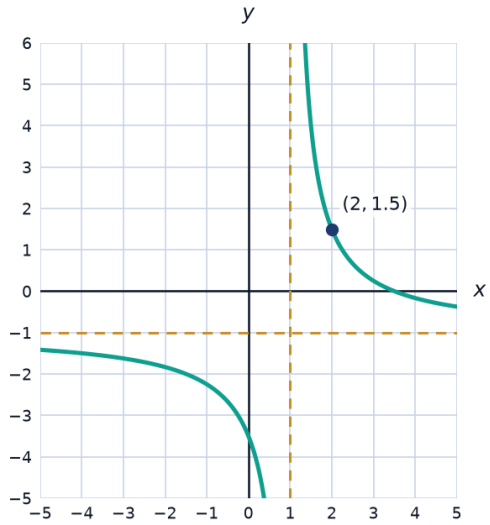
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9 [Intermediate] • Target: Continuity and limits together

A function f is defined by $f(x) = x + 3$ for $x \neq 2$, and $f(2) = 10$. Find $\lim_{x \to 2} f(x)$ and state whether f is continuous at $x = 2$.

10 [Intermediate] • Target: Continuity and limits together

Estimate $\lim_{x \to \infty} \left(\sqrt{x^2 + 1} - x\right)$ by rationalizing.



11 [Intermediate] • Target: Limits from graphs and tables

A function g satisfies $g(1.9) = 5.8$, $g(1.99) = 5.98$, $g(1.999) = 5.998$, $g(2.1) = 6.2$, $g(2.01) = 6.02$, $g(2.001) = 6.002$. Based on this table, estimate $\lim_{x \to 2} g(x)$.

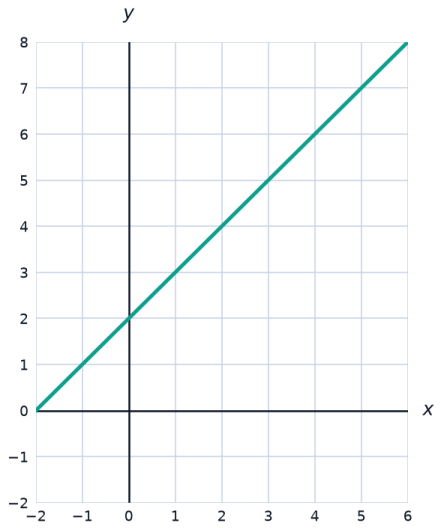
12 [Intermediate] • Target: Limits from graphs and tables

Explain, using one-sided limits, why $\lim_{x \to 0} \frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x \to 0} \frac{1}{x}$ does not exist even as an infinite limit.

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13 **[Intermediate]** • **Target: Visualizing a limit graphically**

The graph of $f(x) = \frac{x^2 - 4}{x - 2}$ (which simplifies to $x + 2$ with a hole at $x = 2$) is shown. Use the graph to estimate $\lim_{x \to 2} f(x)$, and explain why the hole doesn't affect the limit's value.



14 **[Intermediate]** • **Target: The formal idea of a limit (intuitive)**

Explain, in your own words, what it means for $\lim_{x \to a} f(x) = L$, without using formal epsilon-delta language.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Limit laws** • **Cross-Lesson: 4.01 Circles and Parabolas**

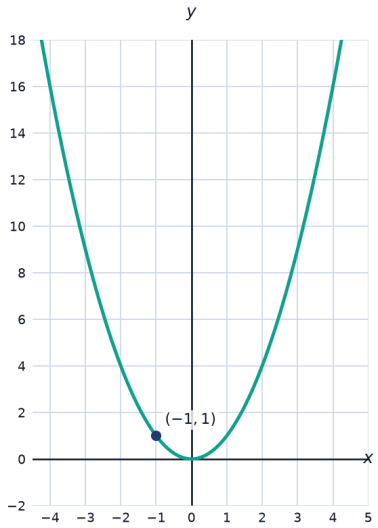
Given $\lim_{x \to 3} f(x) = 5$ and $\lim_{x \to 3} g(x) = -2$, use the limit laws to evaluate:

a $\lim_{x \to 3} [f(x) + g(x)]$ **b** $\lim_{x \to 3} [f(x) \cdot g(x)]$ **c** $\lim_{x \to 3} \frac{f(x)}{g(x)}$

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16 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 4.02 Ellipses

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x \to 1^-} f(x)$, $\lim_{x \to 1^+} f(x)$, and state whether $\lim_{x \to 1} f(x)$ exists.



17 [Challenging] • Target: Limit laws • Cross-Lesson: 4.03 Hyperbolas

On the EST II Math exam, Advanced Algebra and Functions together account for 40% of the questions. In a 50-question test, if a student targets an accuracy rate of 90% in these two high-yield domains, how many questions from these domains can the student afford to miss at most?

18 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 5.01 Vectors: Components and Operations

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

19 [Challenging] • Target: Evaluating limits • Cross-Lesson: 5.02 The Dot Product

If $f(x) = x^2 - 1$ and $g(x) = \sqrt{x + 2}$, find (a) $f(g(3))$, (b) the domain of $g \circ f$.

20 [Challenging] • Target: One-sided limits and limits at infinity • Cross-Lesson: 5.03 Parametric Equations

If $\sin\theta = -\frac{4}{5}$ and θ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating limits

A function is one-to-one if:

a. Every output repeats

b. Passes the vertical line test only

c. Passes the horizontal line test

d. Is a polynomial

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22 [Foundational] • Target: One-sided limits and limits at infinity

[Transformation Trial 1] Identify the matrix representing a counterclockwise rotation by 360° in $\mathbb{R}^2$.

a. $(-1, 0; 0, 1)$

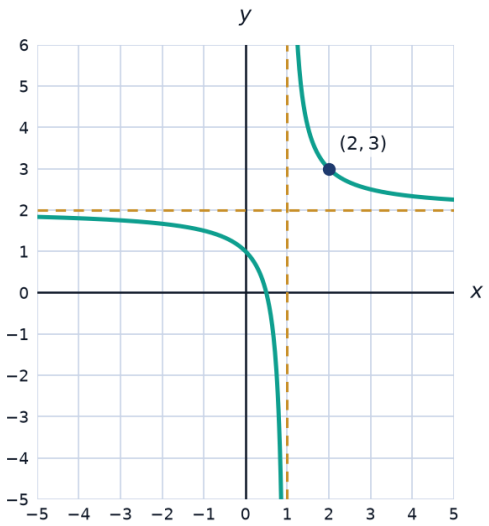
b. $(0, 1; 1, 0)$

c. $(1, 0; 0, 1)$

d. $(1, -1; 1, 1)$

23 [Foundational] • Target: More limit computations

Eliminate the parameter to find y as a function of x: $x = 1t - 2$, $y = -2t^2$.



a. $y = -1(x + 2)^2$

b. $y = -2(x + 2)^2$

c. $y = 2(x + 2)^2$

d. $y = -2(x - 2)^2$

24 [Foundational] • Target: Continuity and limits together

Find the component form of vector $\mathbf{u}$ directed from $(9, 12)$ to $(20, -2)$.

a. $\langle 12, -14 \rangle$

b. $\langle -11, 14 \rangle$

c. $\langle 11, -14 \rangle$

d. $\langle 11, 14 \rangle$

25 [Foundational] • Target: Limits from graphs and tables

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of x. Find this value.

a. x=0

b. x=-4

c. x=5

d. x=4

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26 [Foundational] • Target: Visualizing a limit graphically

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. -14

d. -13

27 [Foundational] • Target: The formal idea of a limit (intuitive)

Let $f(x) = 3x + 2$ and $g(x) = 1/3x - 2/3$. Are f and g inverse functions?

a. No, f and g are not inverses

b. Yes, f and g are inverses

c. Cannot be determined

d. Only if $x = 0$

28 [Foundational] • Target: Limit laws

The function $f(x) = x - e^x$ has exactly one critical point. Find it.

a. $x = e$

b. $x = 1$

c. $x = -1$

d. $x = 0$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Limits involving piecewise functions

What does multiplicity 3 mean for a root graphically?

a. The graph touches but doesn't cross

b. The root appears 3 times in distinct places

c. The graph has a cusp

d. The graph crosses with an inflection-like shape at the root

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30 [Intermediate] • Target: Evaluating limits

Prove: a polynomial of odd degree with real coefficients has at least one real root.

a. By IVT: $P(x) \to +\infty$ and $P(x) \to -\infty$ (or vice versa) as $x \to \pm\infty$, so it must cross zero

b. By Vieta's formulas

c. By the Rational Root Theorem

d. Not always true

31 [Intermediate] • Target: One-sided limits and limits at infinity

End behaviour of $f(x) = -2x^5$ as $x \to +\infty$:

a. $+\infty$

b. $-\infty$

c. 0

d. 2

32 [Intermediate] • Target: More limit computations

Find the determinant of the 2×2 coefficient matrix $A = (4, 2; 3, 4)$.

a. 22

b. 10

c. 13

d. 6

33 [Intermediate] • Target: Continuity and limits together

Which transformation shifts $f(x)$ right by 3?

a. $f(x + 3)$

b. $f(x - 3)$

c. $f(x) + 3$

d. $f(x) - 3$

34 [Intermediate] • Target: Limits from graphs and tables

If $f : \mathbb{R} \to \mathbb{R}$ is both even and odd, then:

a. $f(x) = 0$ for all x

b. $f(x) = x^2$

c. $f(x) = x$

d. $f(x) = 1$

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35 [Intermediate] • Target: Visualizing a limit graphically

Determine if $f(x) = \frac{x^3}{x^2+1}$ is even, odd, or neither.

a. Neither

b. Constant

c. Odd

d. Even

36 [Intermediate] • Target: The formal idea of a limit (intuitive)

Find $f^{100}(x)$ (100th iterate) of $f(x) = x + 1$.

a. $x + 100$

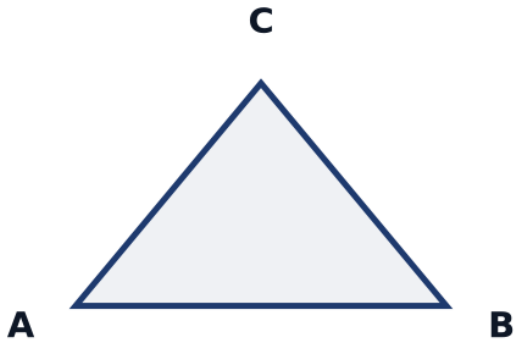
b. $x + 2^{100}$

c. $100x$

d. x^{100}

37 [Intermediate] • Target: Limit laws

$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:



a. $\frac{3}{4}$

b. $\frac{4}{3}$

c. $\frac{3}{5}$

d. $\frac{5}{3}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 5.01 Vectors: Components and Operations

A committee of 4 people is to be selected at random from a group of 10 candidates consisting of 4 mathematicians, 3 physicists, and 3 engineers. Given that the selected committee contains at least one person from each of the three professions, what is the probability that exactly 2 mathematicians are on the committee?

a. $\frac{1}{3}$

b. $\frac{9}{35}$

c. $\frac{3}{7}$

d. $\frac{3}{5}$

39 [Challenging] • Target: Evaluating limits • Cross-Lesson: 5.02 The Dot Product

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f(\frac{\pi}{6}) = 4$. What is the value of $f(\frac{\pi}{3})$?

a. -2

b. 1

c. 4

d. -5

40 [Challenging] • Target: One-sided limits and limits at infinity • Cross-Lesson: 5.03 Parametric Equations

The polynomial $P(x) = 2x^3 + ax^2 + bx - 6$ has a factor of $(x - 2)$. When $P(x)$ is divided by $(x + 1)$, the remainder is -12. What is the remainder when $P(x)$ is divided by $(x - 3)$?

a. 18

b. 24

c. 30

d. 48

41 [Challenging] • Target: More limit computations • Cross-Lesson: 6.01 Polar Coordinates and Conversions

The polynomial $P(x) = x^4 + ax^3 - 7x^2 + bx - 12$ has zeroes at $x = -3$ and $x = 2$, where a and b are real constants. What is the value of $a + b$?

a. -18

b. -2

c. 18

d. 22

42 [Challenging] • Target: Continuity and limits together • Cross-Lesson: 6.02 Polar Equations and Graphs

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

a. 1

b. 2

c. 3

d. 4

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43 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

The function f is defined for $x \in \mathbb{R}, x \neq 3$ by $f(x) = \frac{2x+1}{x-3}$. Which of the following is an expression for the inverse function $f^{-1}(x)$, where $x \neq 2$?

a. $\frac{3x+1}{x-2}$

b. $\frac{x-3}{2x+1}$

c. $\frac{3x-1}{x+2}$

d. $\frac{x+1}{2x-3}$

44 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 7.02 Logarithms and Their Properties

The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

a. $\frac{7}{50}$

b. $\frac{7}{20}$

c. $\frac{7}{13}$

d. $\frac{13}{20}$

45 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

The first three terms of a geometric progression are 32, 24, and 18. Find the sum to infinity, S_∞, of the progression.

a. 128

b. 96

c. 74

d. 64

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